

# Task 7: Identify and Remove Suspicious Browser Extensions

## Objective

The main objective of this task is to learn how to find and remove browser extensions that may be unsafe or not useful. Browser extensions can make work easier, but some of them can be harmful. By checking and removing suspicious extensions, browser security, privacy, and performance can be improved.

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## Tools Used

- Google Chrome web browser
  - Chrome Extensions Manager
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## Steps Taken

1. Opened the Google Chrome browser on the computer.
2. Clicked on the three-dot menu (:) in the top-right corner of the browser.
3. Selected **Extensions** and then clicked on **Manage Extensions**.
4. A list of all installed browser extensions was displayed.
5. Reviewed each extension carefully, one at a time.
6. For every extension, the following points were checked:
  - The name of the extension and the developer who created it
  - The permissions requested by the extension, such as access to websites or browsing data

- User reviews and ratings on the Chrome Web Store
  - Whether the extension was still useful or used regularly
7. Identified extensions that showed signs of being suspicious, including those that:
- Were not used anymore
  - Had low ratings or many negative reviews
  - Requested too many permissions for their function
  - Were installed automatically or without clear knowledge
8. Removed all extensions that were not needed or seemed unsafe.
9. Restarted the browser to apply the changes.
10. Checked the browser again to see if performance or behavior improved.

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## Suspicious / Unnecessary Extensions Found

| Extension Name              | Reason for Suspicion                                                              | Action Taken |
|-----------------------------|-----------------------------------------------------------------------------------|--------------|
| Free PDF Converter          | Requested access to all websites and browsing data even though it was rarely used | Removed      |
| Coupon Finder Plus          | Showed unwanted ads and pop-ups and had many negative reviews                     | Removed      |
| Unknown Extension (No Icon) | Installed without remembering when or why and had no clear purpose                | Removed      |

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## Results After Removal

After removing the suspicious and unnecessary extensions, several improvements were noticed:

- The browser started faster than before
  - Unwanted pop-ups and advertisements were reduced
  - Browsing felt smoother and more responsive
  - No important browser features were lost
  - The browser felt safer and cleaner to use
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## How Malicious Extensions Can Harm Users

Some browser extensions can be dangerous if they are created with bad intentions or are poorly designed. These extensions can:

- Track user browsing activity without permission
- Collect personal or sensitive information
- Show unwanted advertisements on websites
- Change search results or redirect users to unsafe websites
- Slow down browser and system performance
- Be used to steal information or perform phishing attacks

Because of these risks, it is important to only install trusted extensions and review them regularly.

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## ExtAnalysis

**ExtAnalysis** is a Python-based security framework designed for the static and dynamic analysis of web browser extensions (Chrome, Firefox, Brave, and Opera). It allows security researchers to "peek under the hood" of extensions to find malicious code, hidden trackers, or vulnerabilities.

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### Key Features

- **Automatic Downloader:** Fetches extensions directly from the Chrome Web Store or Firefox Addons site.
- **Manifest Viewer:** Analyzes manifest.json to identify dangerous permissions (e.g., access to all URLs, cookies, or tabs).
- **Intel Extraction:** Automatically scrapes files for:
  - URLs, Domains, and IP addresses.
  - Bitcoin and Email addresses.
  - Base64 encoded strings and developer comments.
- **Vulnerability Scanning:** Integrates with RetireJS to find outdated/vulnerable JavaScript libraries and VirusTotal for threat intelligence.
- **Visual Graphing:** Creates a network graph showing the relationship between all files and the external URLs they contact.

## Conclusion

This task helped in understanding the importance of managing browser extensions. Not all extensions are safe or useful, and some can cause security and privacy problems. By regularly reviewing and removing suspicious or unused extensions, users can protect their personal information and improve browser performance. This is a simple but effective way to keep a system safe and secure.